

Computer Sciences I

Semester 2026-III

2-Player Videogame Project Assignment

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Team Project Assignment

Each team is assigned one unique two-player videogame where one player is controlled by the computer using a simple AI algorithm. All games share the same technical architecture (Python with game logic, AI opponent, and Pygame UI) and are evaluated with the same rubric. Only the game mechanics and theme differ, ensuring a fair workload across all teams.

Teams are fixed for the entire semester. Role distribution within each team (Game Logic Developer, AI Algorithm Developer, UI Developer) is decided internally and must be declared in the **Checkpoint document** due at Week 8.

Team and Game Distribution

Team	Game	AI Strategy	Students
1	Tic-Tac-Toe	Minimax algorithm to find optimal moves and block player threats.	Avila Tafur Juan Diego Cipagarua Coronado Sergio Andres Marino Figueroa Natalie
2	Rock Paper Scissors	Pattern recognition from player history with weighted randomization.	Rodriguez Duarte Irene Santiago Lozano Nieto David Alejandro Pardo Delgado Luis Manuel

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Any comment or concern about this document can be sent to Carlos A. Sierra at: *cavir-guezs@udistrital.edu.co*.

3	Connect Four	Minimax with alpha-beta pruning to prevent chains and create winning opportunities.	Mora Barrera Juan Camilo Peñaloza Quintero Ivan Andres Campos Alvarado Julian David
4	Checkers	Greedy piece evaluation with lookahead to maximize captures and board control.	Cubillos Gomez Jeronimo Macias Acevedo Kevin Jair Vargas Huertas Juan Manuel
5	Memory Game	Perfect recall strategy to match all pairs with optimal efficiency.	Nossa Orjuela Isabella Herrero Cifuentes Delan Santiago Cusba Marin Gabriel Mateo
6	Snake Battle	Path planning with greedy food targeting and collision avoidance for two snakes.	Bonilla Piñeros Brayan Esteban Zibieta Rincon Luis Felipe Barrantes Alvarado Suhayl Mariana Quintero Puentes Miguel Angel
7	Pong	Predictive paddle positioning based on ball trajectory and velocity.	Fajardo Gomez Daniel Andres Velasquez Leguizamo Juan Sebastian Ruiz Fernandez Julio Lozano Moreno Davinson Sfrey

Game Details

The following descriptions provide each team with a clear starting point for their game design.

1. **Tic-Tac-Toe:** The classic 3×3 grid game. The player places marks (X) against the computer opponent (O). The AI uses the minimax algorithm to evaluate all possible future states and select the move that maximizes its chances of winning or forces a draw. Implementation includes board state representation, win-condition checking, and recursive minimax search with early termination.
2. **Rock Paper Scissors:** A best-of-five hand game where both players simultaneously choose. The player inputs their choice while the computer selects based on observed patterns in the player's recent choices. The AI maintains a statistical model of player behavior and applies weighted randomization to make informed predictions. Implementation includes pattern matching algorithms and probability-based decision making.
3. **Connect Four:** A 7×6 grid where players drop pieces to create four in a row. The player uses red pieces and the computer uses yellow. The AI uses minimax with alpha-beta pruning to limit search depth while still playing strategically. Implementation includes column-based piece mechanics, gravity simulation, and efficient board evaluation.

4. **Checkers:** A simplified checkers game on an 8×8 board with standard rules (diagonal movement, capturing opponent pieces, king promotion). The AI uses greedy evaluation based on piece count and position value, with look-ahead to anticipate consequences of moves. Implementation includes piece movement validation, capture detection, and board state scoring.
5. **Memory Game:** A grid-based matching game with hidden card pairs. Players take turns flipping two cards trying to find matches. The computer opponent plays perfectly—it remembers every revealed card and always makes optimal choices. Implementation includes card tracking, match validation, and turn-based game flow.
6. **Snake Battle:** Two snakes move simultaneously on a grid trying to eat food and avoid collisions with walls and each other. One snake is controlled by the player, one by the computer. The AI snake uses path-planning to locate food while avoiding obstacles, employing greedy nearest-target selection. Implementation includes collision detection, food spawning, and independent AI pathfinding.
7. **Pong:** The classic arcade game on a centered court. The player controls one paddle, the computer controls the other. A ball bounces between them. The AI paddle predicts the ball's trajectory based on velocity and position, positioning itself to intercept. Implementation includes physics-based ball movement, collision detection with paddles, and predictive paddle positioning.

Checkpoint and Final Submission Summary

Milestone	Due	Deliverable
Checkpoint	Week 8	PDF with game rules, AI algorithm pseudocode, class diagrams, Pygame wireframe, and role assignment statement. No code required.
Final Submission	Week 16	Working game with AI opponent, GitHub repository, technical report (PDF), and live in-class demo.

Game assignments were made in order of student list. All games have been designed to require equivalent effort for AI implementation, game mechanics, and user interface. If your team has questions about your assigned game, please attend office hours or post in the course forum.